AI 101 for Mechanical Engineers · Module 6

Baselines prevent empty success claims

Lesson 6.3

What you will learn

> Interpret a performance result using a fair baseline, identify the context needed for an error metric, and explain what a random seed helps reproduce.

Lesson overview

Module 6 · Lesson 3

Baselines prevent empty success claims

AI 101 for Mechanical Engineers

1 Compare with a simple baseline

A baseline is a simple comparison approach.

Regression	Classification
Predict the training mean, or use a trusted physical relationship.	Use a majority-class rule, with suitable fault-sensitive metrics.

Keep the comparison fair

- ✓ Use the same evaluation examples.
- ✓ Use comparable information available at prediction time.
- ✓ State the metric.

A training score alone is not a fair held-out comparison.

2 Read the MAE comparison

MAE is the average absolute difference between predictions and reference outputs. It has the same unit as the predicted output.

Method	MAE on the same held-out tests
ML model	1.2 kW = 1,200 W
Physical baseline	0.8 kW = 800 W

The baseline has the lower MAE.
The ML model has not shown an improvement on this metric.

If the comparison is sound, using the baseline is a valid engineering decision. Cost, interpretability and operating range also matter.

3 Interpretable, repeatable results

Record the context

- ✓ Target quantity and unit
- ✓ Evaluated examples and operating conditions
- ✓ Data split and reference uncertainty
- ✓ Baseline result, preprocessing and model version

What does a random seed do?

It helps repeat a random split or training procedure under stated conditions. It does not prove accuracy or make biased data representative.

Different software, hardware or parallel execution may still change results.

Key takeaway — **Extra complexity needs evidence of a benefit. Report a stronger baseline honestly.**

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Choose a simple baseline for comparison

Task	Baseline
Regression	Training mean or trusted physical relationship
Classification	Majority-class rule, with fault-sensitive metrics

Same evaluation examples, comparable information, stated metric. A training score alone is not a fair held-out comparison.

MAE keeps the unit of the output

- Average absolute difference between predictions and reference outputs
- Same unit as the output being predicted

The physical baseline has the lower MAE

ML model

- MAE 1.2 kW (1,200 W)

Physical baseline

- MAE 0.8 kW (800 W)
- Same held-out tests

> The more elaborate model has not demonstrated an improvement on that metric.

One metric does not settle every consideration

- Cost, interpretability and operating-range evidence also matter
- Saying a system uses AI is not evidence of benefit

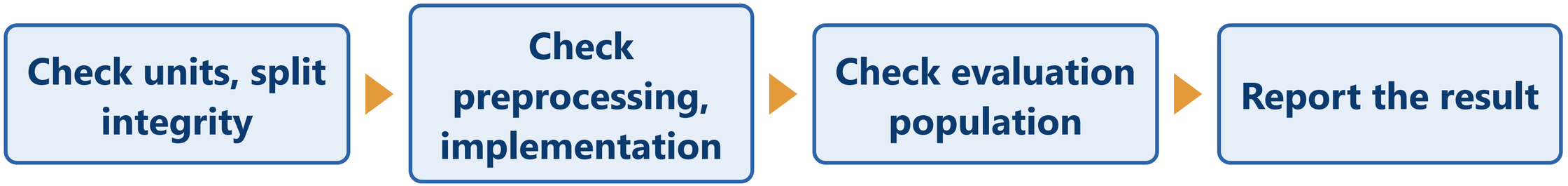
Context needed to interpret an error value

1. Target quantity and unit
2. Evaluated population and operating conditions
3. Data split and which examples informed each stage
4. Reference measurement uncertainty
5. Baseline error on the same evaluation examples

A seed supports repeatability, not guaranteed validity

- A random seed initializes a pseudorandom procedure [36]
- Helps repeat a split or training under stated conditions
- Does not prove accuracy or eliminate uncertainty
- Does not make a biased dataset representative

If the baseline wins, check, then report



- If sound and the baseline stays better, using it is valid
- Complex models should earn extra maintenance and validation burden with evidence

Record this before sharing a trained model

- Intended task, operating domain, data source and date range
- Split procedure, preprocessing steps and model settings
- Evaluation results, known limitations, software versions, random seed
- Seeds do not guarantee identical results across software, hardware or parallel execution

CHECK YOUR UNDERSTANDING

What information is needed to interpret “our MAE is 1.2 kW”?

Answer before moving on.

CHECK YOUR UNDERSTANDING

What information is needed to interpret “our MAE is 1.2 kW”?

> Identify the target, evaluated examples and conditions, data split, reference uncertainty, and a fair baseline result. The supplied baseline has MAE 0.8 kW on the same tests.

CHECK YOUR UNDERSTANDING

Does using a random seed guarantee a representative dataset?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Does using a random seed guarantee a representative dataset?

> No. It helps repeat the specified random procedure under stated conditions; representativeness needs separate evidence.

Key takeaways

- Compare with a simple baseline on common examples and metrics
- Interpret error magnitudes with units and evaluation context
- Document the data split, preprocessing and model version
- A random seed supports repeatability, not a guarantee of validity

References

[36] J. Pineau et al., “Improving reproducibility in machine learning research (A report from the NeurIPS 2019 reproducibility program),” J. Mach. Learn. Res., vol. 22, no. 164, pp. 1–20, 2021.

Numbers match the course reference list. The written lesson is the authoritative text.